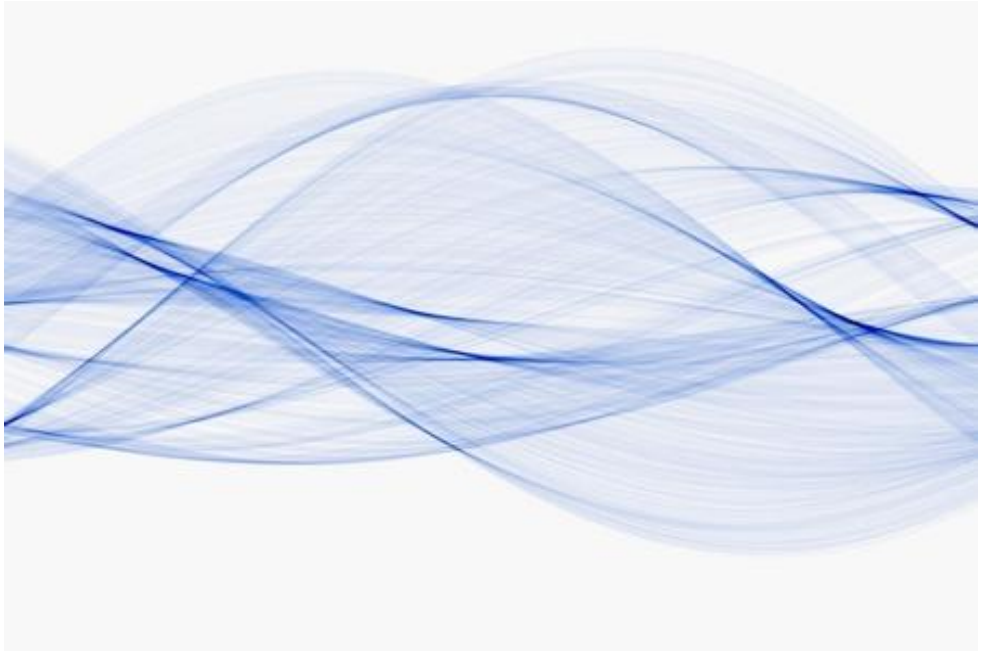
Polarization



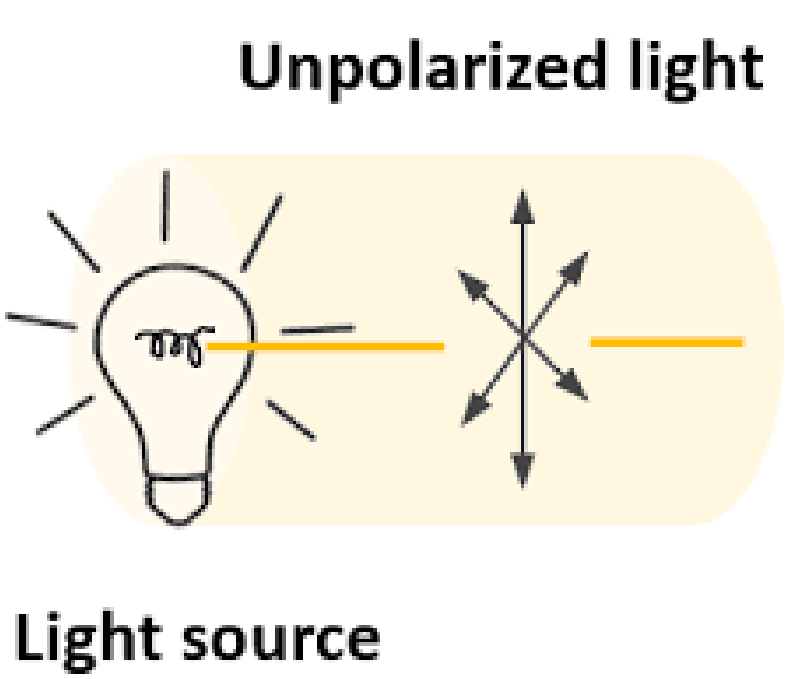
Dr. Monir Morshed
Professor
Dept. of ICT, MBSTU

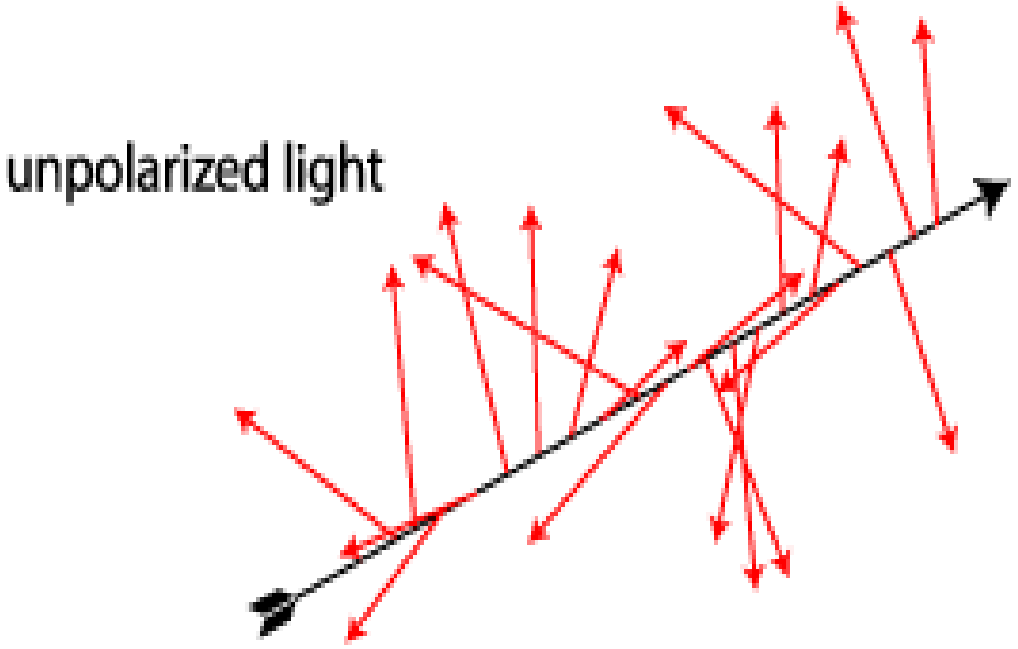
Polarization

- **Polarization** of a plane wave is the orientation of the oscillations of the ***E*** field; perpendicular to the direction of propagation
- For a simple harmonic wave, the electric vector in orthogonal directions may have:
  - Different amplitude
  - Different phase
- The resulting wave is
  - Linearly, elliptically or circularly polarized

EM Waves

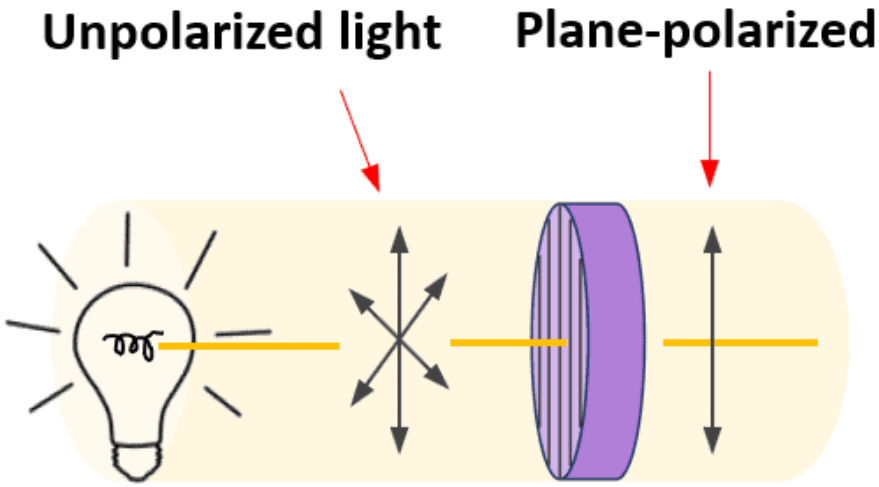
- Light is an electromagnetic wave. EM waves are transverse. Thus, the electrical field can vibrate in any direction perpendicular to the direction of propagation. Most light sources (candles, incandescent light bulbs etc..) emit light that is unpolarized – the electric field has all possible directions of vibrations.





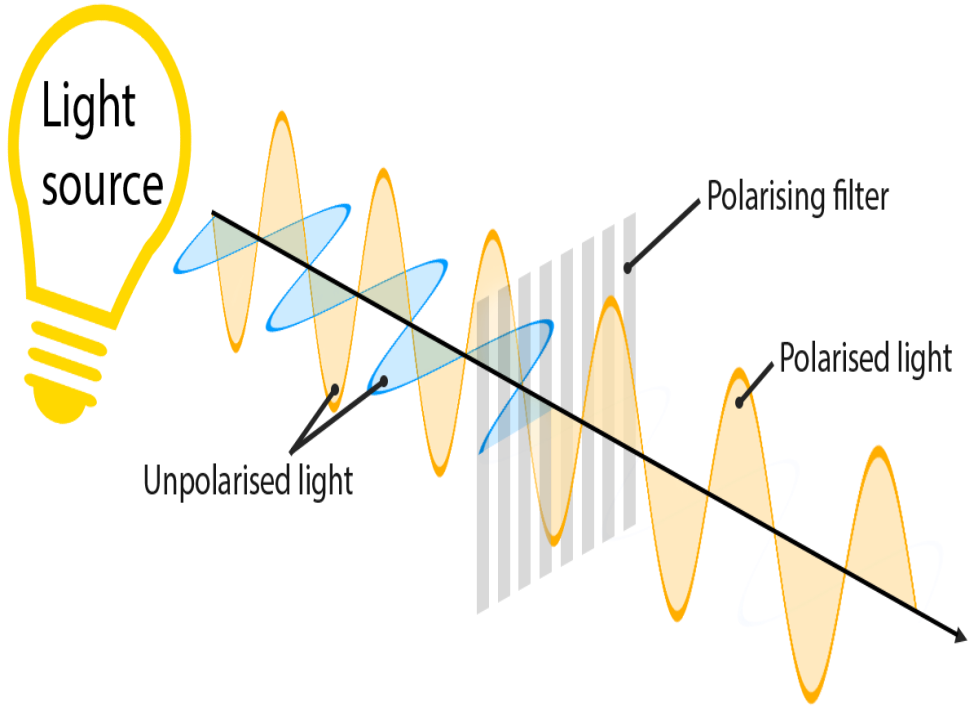
EM Waves

- However it is possible to have polarized light. Polarized light occurs when the vibrations of the electrical field are confined to one plane.



Light source

Polarizer - passes the light in *only one plane*.



Polarization states

- If the resulting electric field is oscillating along a straight line, it is called a linearly polarized (LP) or plane polarized wave
- If the ***E*** field rotates in a circle (constant magnitude) it is called circularly polarized
- If the *E* field rotates in an ellipse then, it is called elliptically polarized wave
- Natural light has random polarization – The orientation of ***E*** filed keep changing randomly

Polarization states

- The **E field and H field of an EM wave are always *orthogonal*, *either*** component completely determines the other.
- An EM wave propagates by exchanging energy between its **E field and H field (analogous to an LC oscillator). If E field is at the maximum, the** H field will vanish (and vice versa).
- There are two independent polarizations for each *monochromatic wave* with the same wave vector ***k because of the two spatial dimensions (x*** and y). When the phase difference between the two polarizations is

  0^0 - the EM wave is *plane-polarized*

  90^0 - the EM wave is *circularly polarized*

  arbitrary - the EM wave is *elliptically polarized*

- *Reflection and refraction occur at the boundary interfaces - a result of* boundary condition for EM field.

Polarization of Light

Let us consider, time evolution of projections of the electric field on two orthogonal axis with

$$\vec{E}(x,y) = E_x\vec{x} + E_y\vec{y}$$
$$E_x = a_x\cos(\omega t - \beta z)$$
$$E_y = a_y\cos(\omega t - \beta z + \ )$$

Let $A = \omega t - \beta z$

$$E_x = a_x \cos A$$
$$E_y = a_y\cos(A+\delta)$$

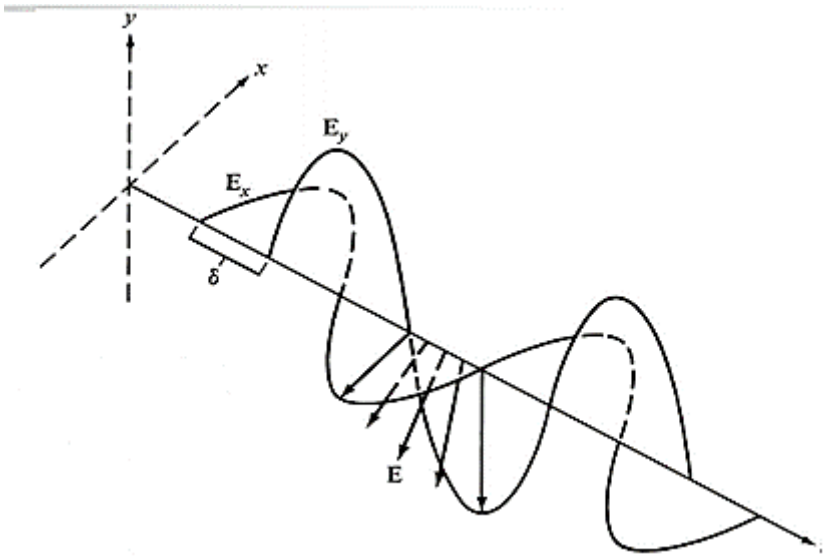
$$\Rightarrow \frac{E_x}{a_x} = cosA$$

$$\frac{E_y}{a_y} = cosAcos\delta - sinAsin\delta$$

$$= \frac{E_x}{a_x}\cos\delta - \sqrt{1-\left(\frac{E_x}{a_x}\right)^2}\,sin\delta$$

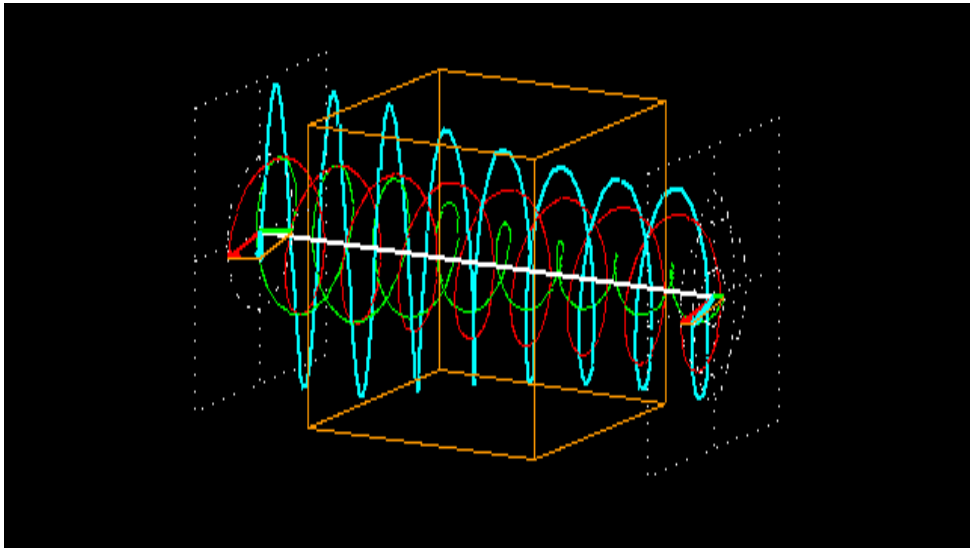
$$\left(\frac{E_x}{a_x}\cos\delta - \frac{E_y}{a_y}\right)^2 = \left[1-\left(\frac{E_x}{a_x}\right)^2\right]sin^2\delta$$

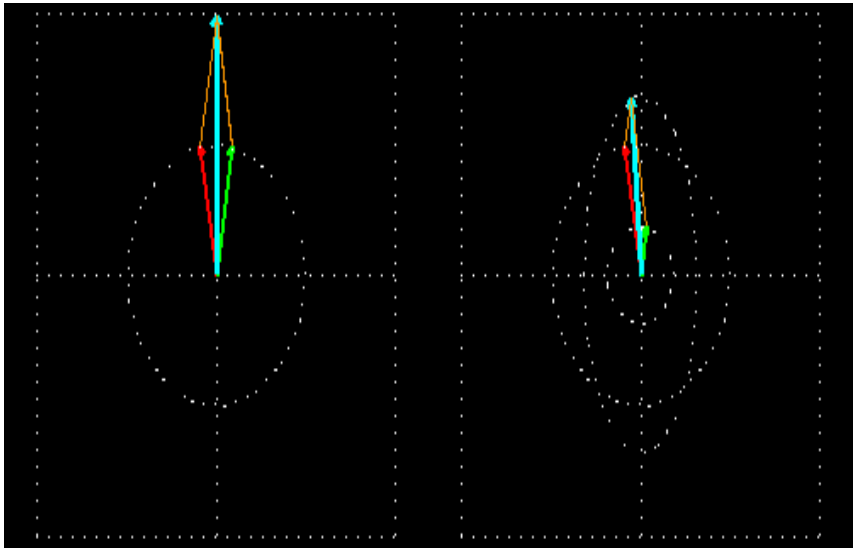
$$\left(\frac{E_x}{a_x}\right)^2 + \left(\frac{E_y}{a_y}\right)^2 - 2\frac{E_x}{a_x}\frac{E_y}{a_y}cos\delta = sin^2\delta$$ Polarization Ellipse.



Elliptical polarization

- The two components are not in phase and either do not have the same amplitude and/or are not 90 degrees out of phase

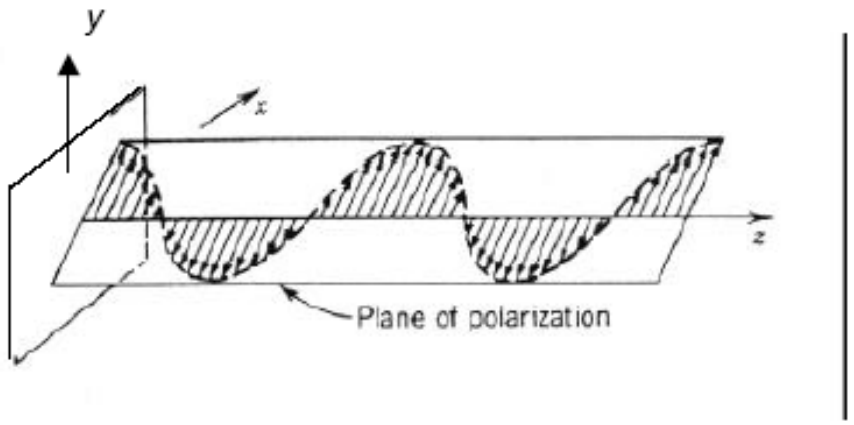


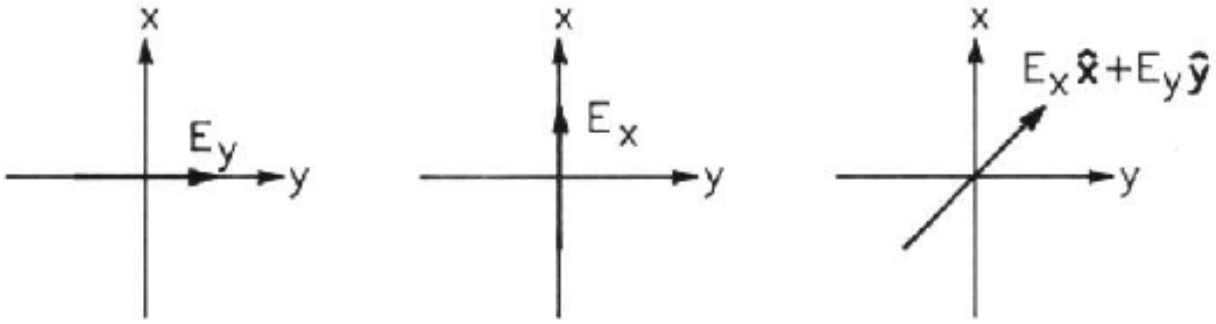


Linear Polarization

LINEARLY polarized light

- one of the components E_x or E_y = 0
- if ϕ = 0 or π then $\tilde{E}_y = \frac{E_y}{E_x}\tilde{E}_x$
- angle is determined by E_y and E_x

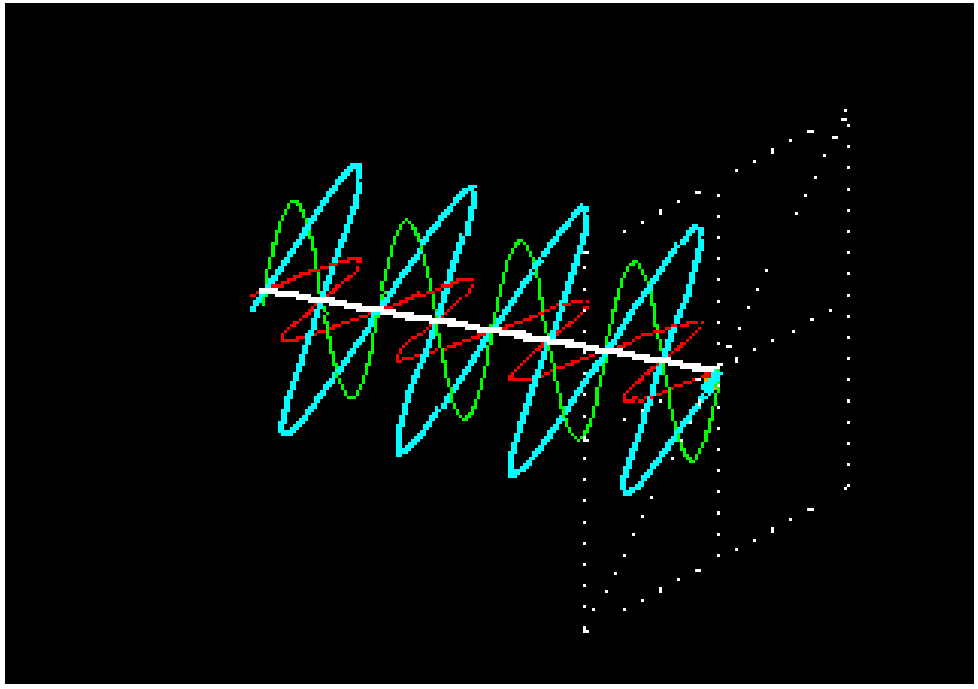


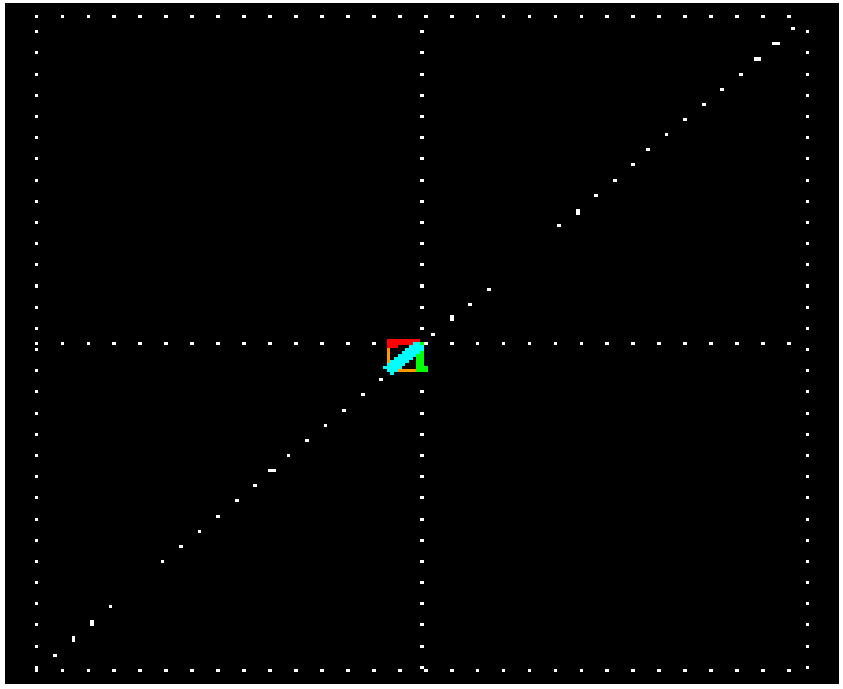


horizontal vertical general

Linear polarization

- The two orthogonal components are in phase



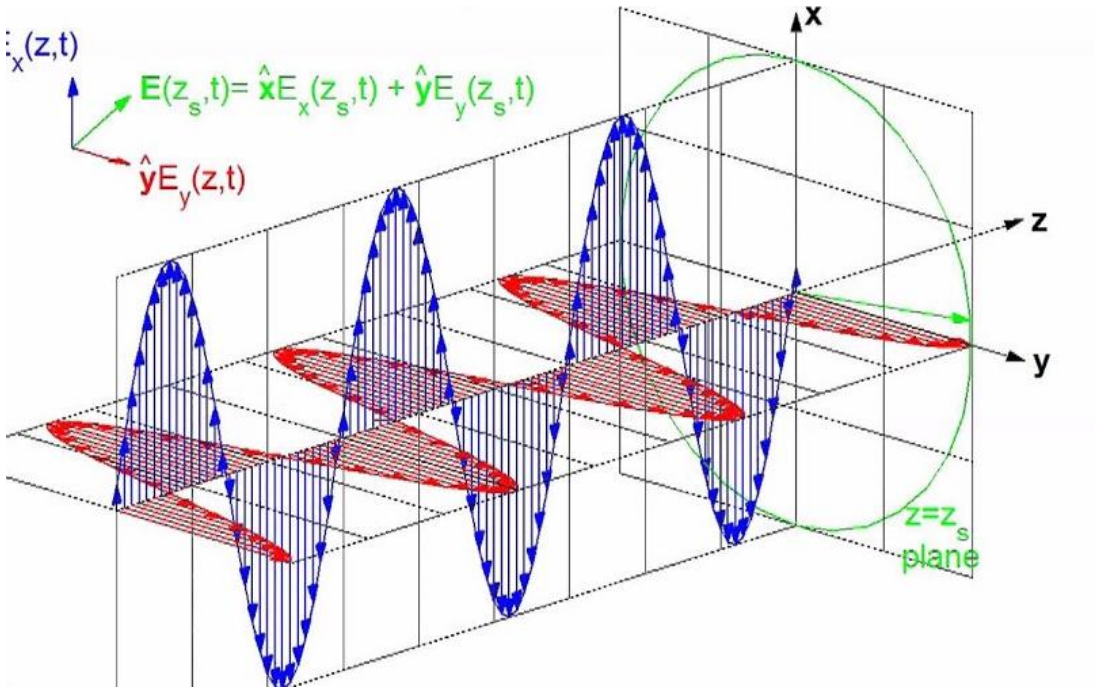


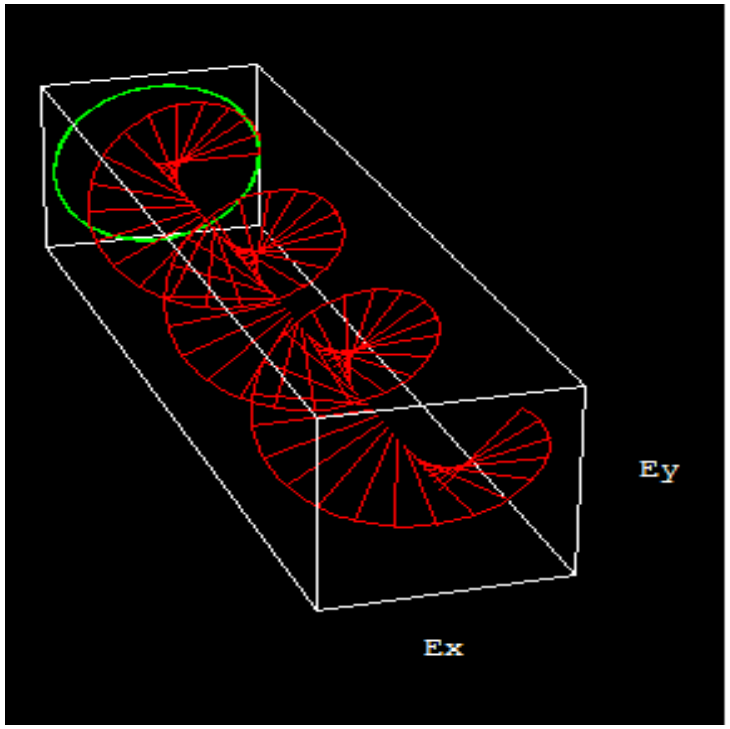
Circular Polarization

Polarization Ellipse,

$$\left(\frac{E_x}{a_x}\right)^2+\left(\frac{E_y}{a_y}\right)^2-2\frac{E_x}{a_x}\frac{E_y}{a_y}cos\delta=sin^2\delta$$

Consider, when $\delta = {}^{\pi}/_{2}$, $\left(\frac{E_x}{a_x}\right)^2+\left(\frac{E_y}{a_y}\right)^2$=1, If a_x=a_y=a, the locus of the resultant E(x,y) will be a circle.

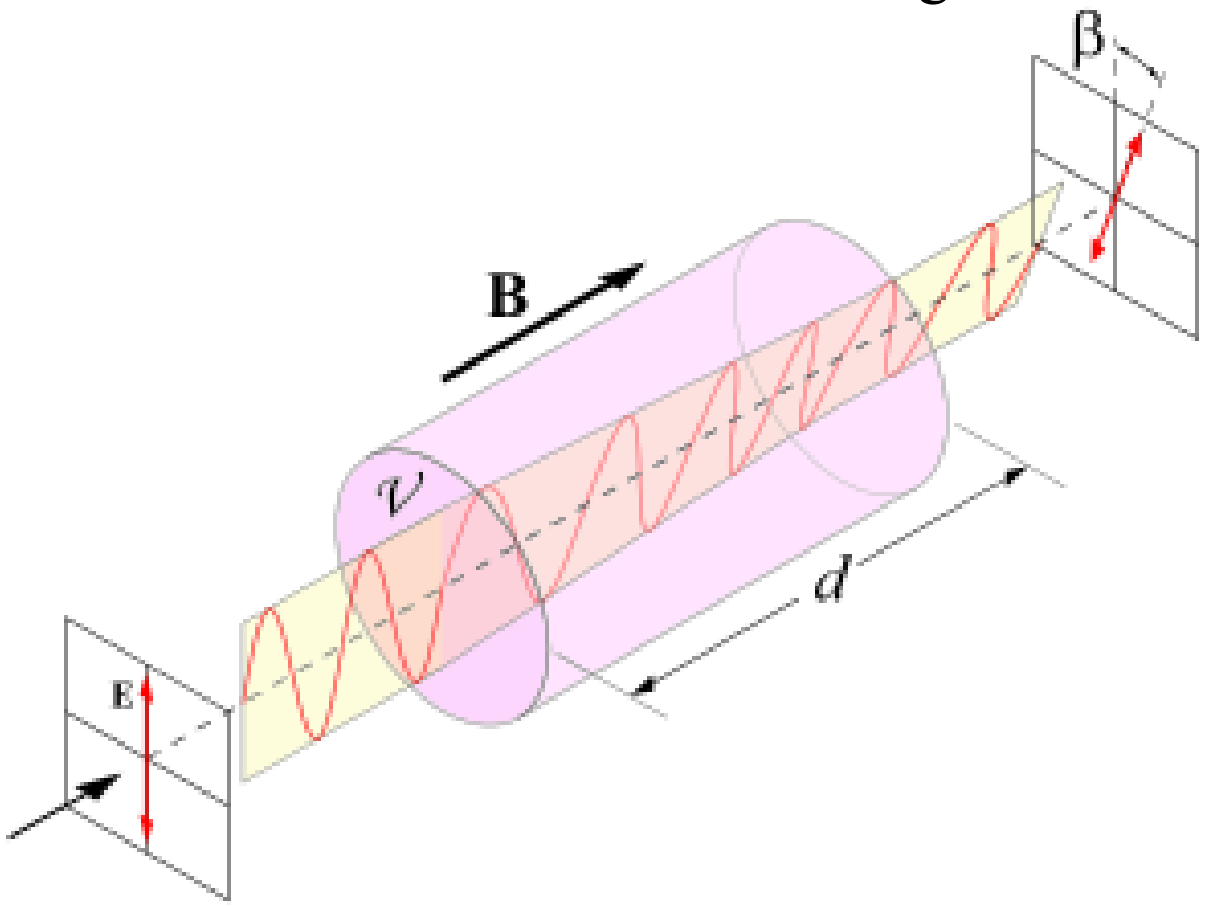




- the two orthogonal components have same amplitude and are +/-90 degrees out of phase.

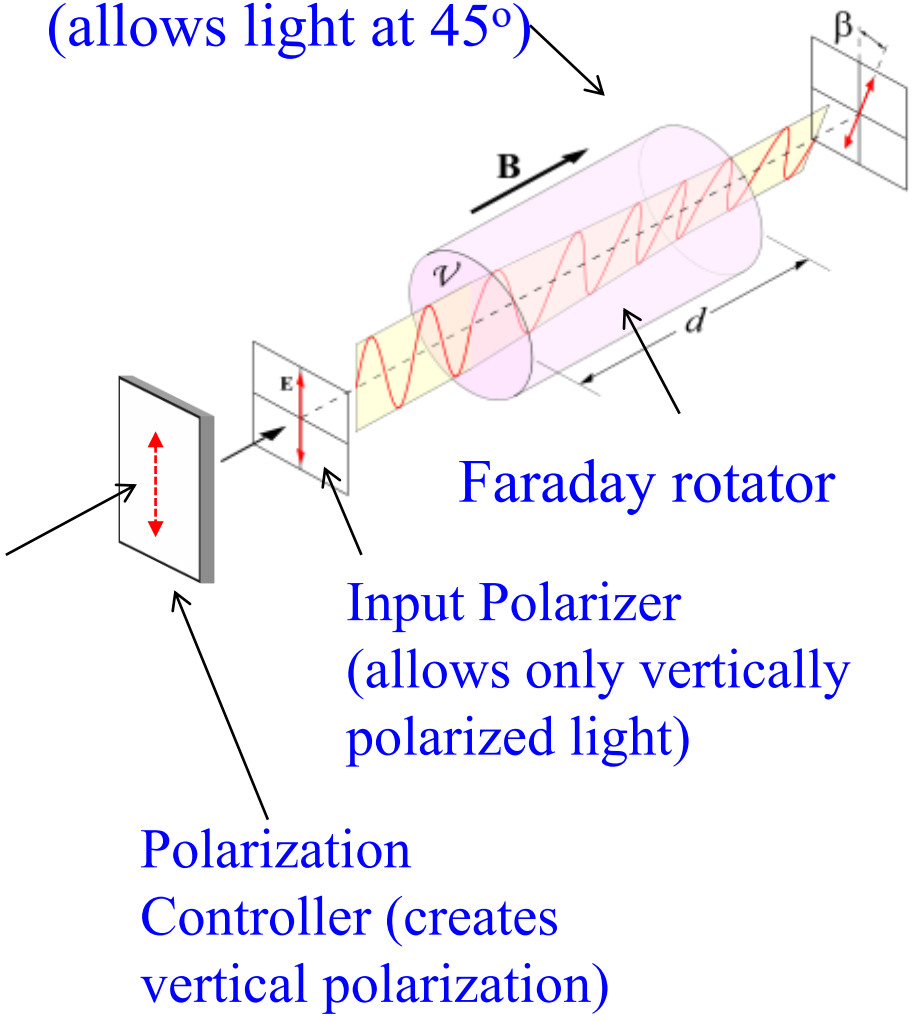
Faraday Effect

- When a magnetic field is applied to linearly polarized light, the plane of polarization rotates.
- The rotation is proportional to the intensity of the applied magnetic field in the direction of the beam of light



This effect is used in Optical Isolators

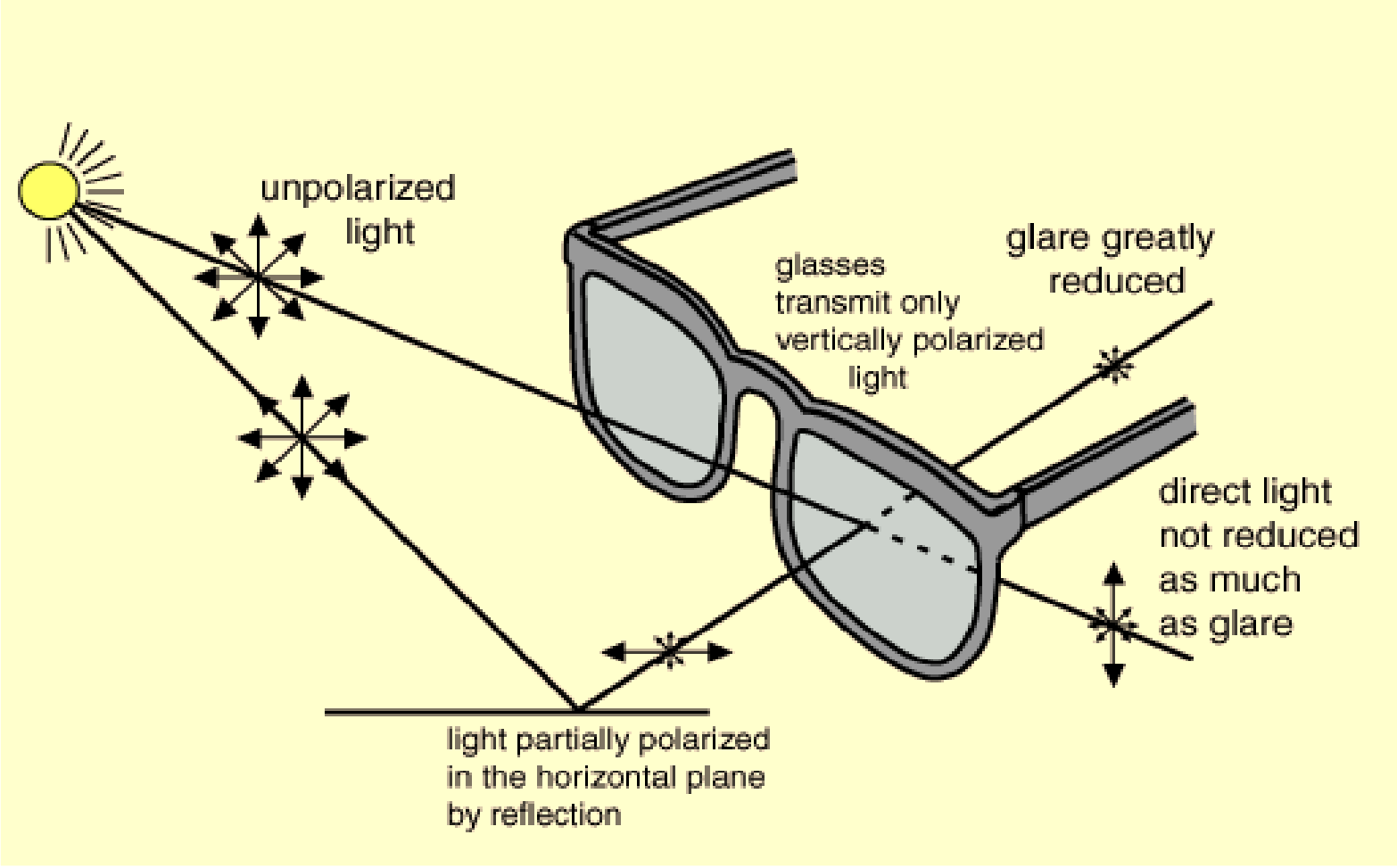
Optical Isolator



- Vertically polarized light enters the isolator.
- The Faraday rotator rotates it by 45°.
- Output polarizer passes the light.
- Backward traveling (reflected) light starts with 45° tilt.
- It gets horizontal polarization at the rotator and will be extinguished.

Applications

- Knowing that reflected light or glare from surfaces is at least partially plane polarized, one can use Polaroid sunglasses. The polarization axes of the lenses are vertical as the glare usually comes from reflection off horizontal surfaces.









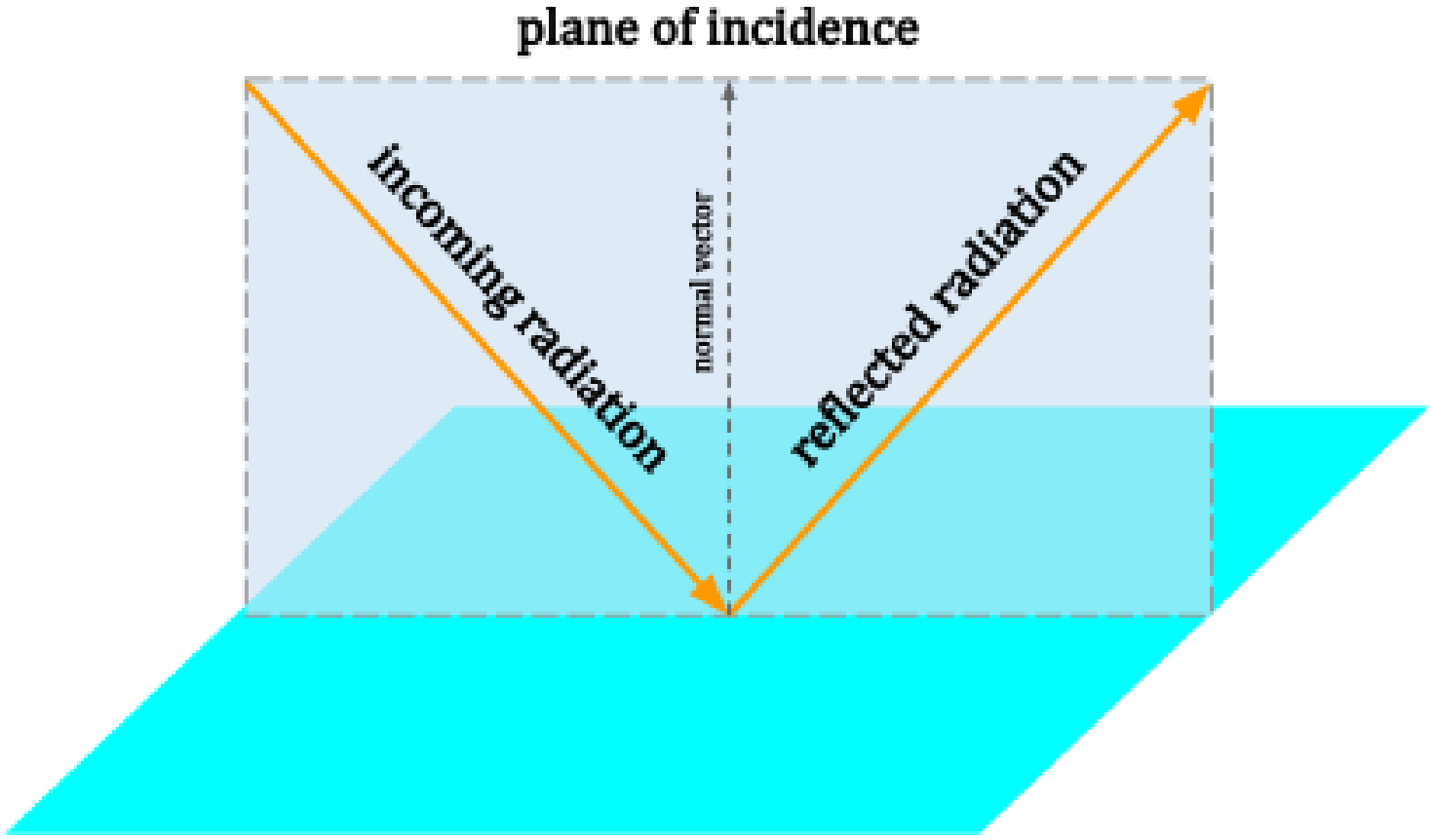
Without polarization



With polarization

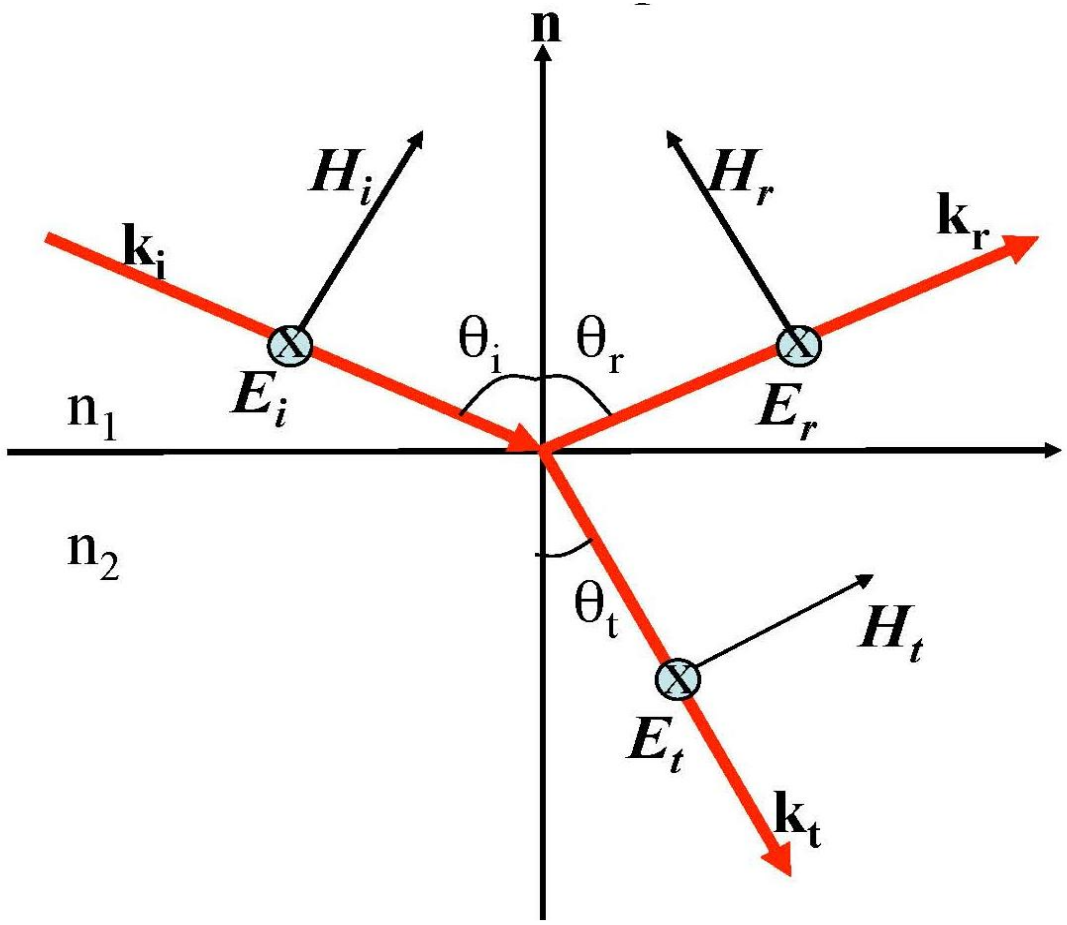
Plane of incidence

- The plane which contains the incoming radiation's surface normal and propagation vector is called plane of incidence. It is also called the incidence plane or the meridional plane.



TE polarization

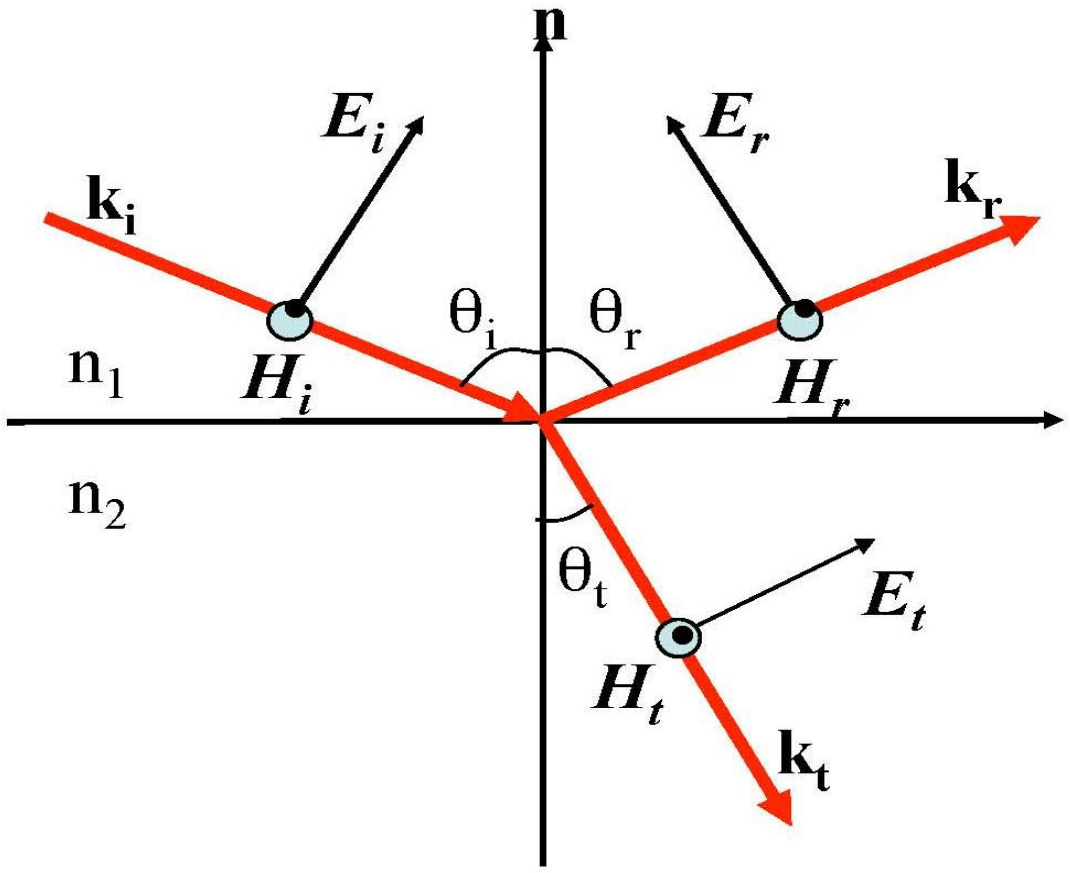
- The electric field is linearly polarized in a direction perpendicular to the plane of incidence, while the magnetic field is polarized parallel to the plane of incidence. This is called the transverse electric (TE) polarization. This wave is also called s-polarized.



S-senkrecht

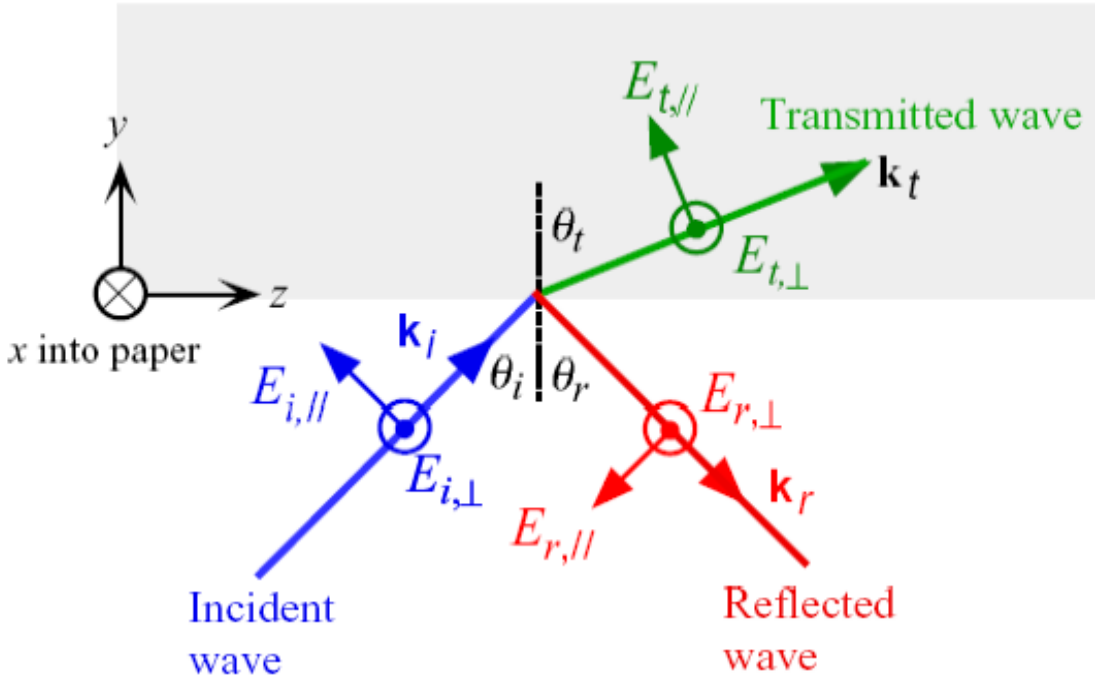
TM polarization

- The electric field is linearly polarized in a direction parallel to the plane of incidence, while the magnetic field is polarized perpendicular to the plane of incidence. This is called the transverse magnetic (TM) polarization. This wave is also called p-polarized.



S-senkrecht

Fresnel's Equations



The **incidence plane** is the plane containing the incident and reflected light rays.

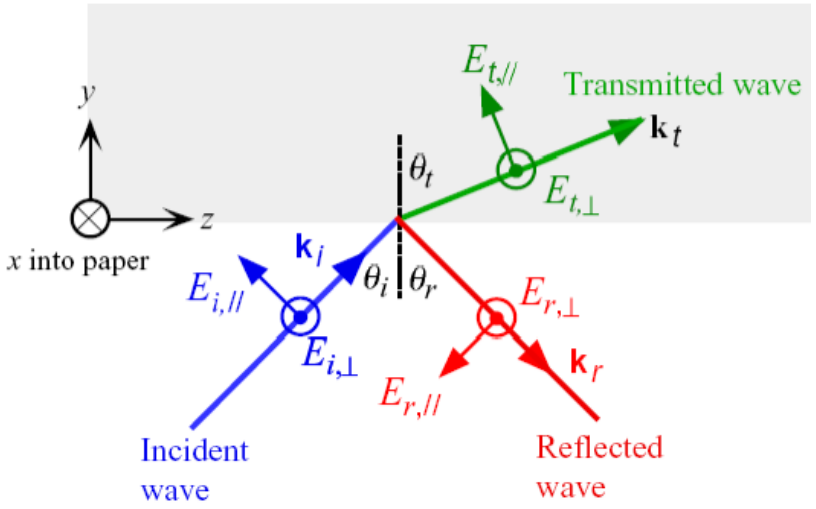


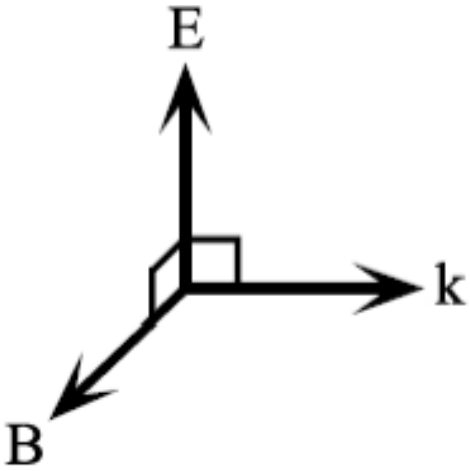
Augustin Fresnel (1788-1827)

Transverse electric field (TE) waves: $E_{i,\perp}$, $E_{r,\perp}$, and $E_{t,\perp}$.
Transverse magnetic field (TM) waves: $E_{i,//}$, $E_{r,//}$, and $E_{t,//}$.

$$E_i = E_{i0} \exp j(\omega t - \mathrm{k}_i \cdot \mathrm{r})$$
$$E_r = E_{r0} \exp j(\omega t - \mathrm{k}_r \cdot \mathrm{r})$$
$$E_t = E_{t0} \exp j(\omega t - \mathrm{k}_t \cdot \mathrm{r})$$

Our goal is to find E_{r0} and E_{t0} relative to E_{i0}, including any phase changes

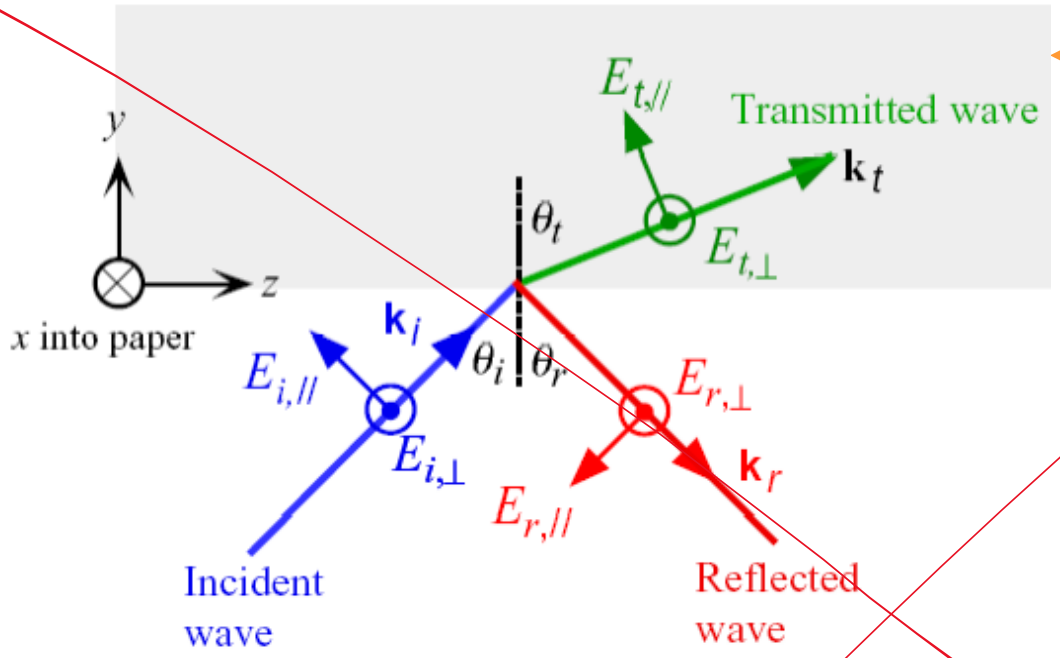


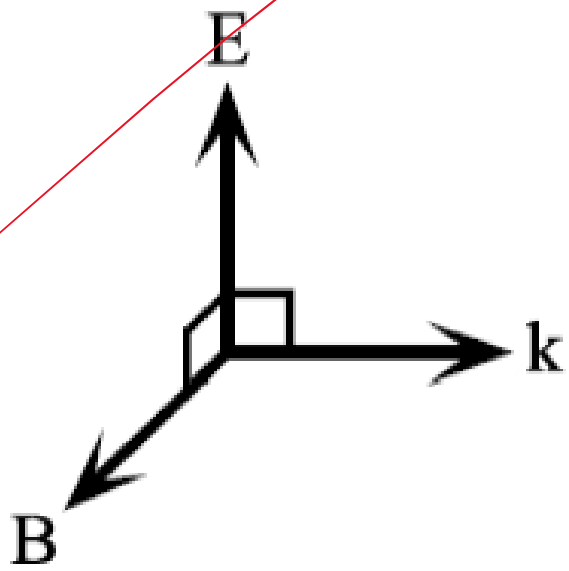


$$\theta_i = \theta_r$$
$$n_1 \sin\theta_i = n_2 \sin\theta_t$$
} Snell's law

$$B_\perp = (n/c)E_{//}$$
$$B_{//} = (n/c)E_\perp$$
} Maxwell's equations

$$E_{1,\text{tangential}} = E_{2,\text{tangential}}$$
$$B_{1,\text{tangential}} = B_{2,\text{tangential}}$$ (for non-magnetic media)
} Boundary conditions





$$-E_{\mathrm{i},\perp}-E_{\mathrm{r},\perp}=-E_{\mathrm{t},\perp}$$

$$-B_{\mathrm{i},\perp}+B_{\mathrm{r},\perp}=-B_{\mathrm{t},\perp}\Rightarrow-(n_1/c)E_{\mathrm{i},//}+(n_1/c)E_{\mathrm{r},//}=-(n_2/c)E_{\mathrm{t},//}$$

$$-E_{\mathrm{i},//}\cos\theta_{\mathrm{i}}-E_{\mathrm{r},//}\cos\theta_{\mathrm{r}}=-E_{\mathrm{t},//}\cos\theta_{\mathrm{t}}$$

$$B_{\mathrm{i},//}\cos\theta_{\mathrm{i}}-B_{\mathrm{r},//}\cos\theta_{\mathrm{r}}=B_{\mathrm{t},//}\cos\theta_{\mathrm{t}}\Rightarrow$$

$$(n_1/c)E_{\mathrm{i},\perp}\cos\theta_{\mathrm{i}}-(n_1/c)E_{\mathrm{r},\perp}\cos\theta_{\mathrm{r}}=(n_2/c)E_{\mathrm{t},\perp}\cos\theta_{\mathrm{t}}$$

$$\begin{cases} -E_{\mathrm{i},\perp} - E_{\mathrm{r},\perp} = -E_{\mathrm{t},\perp} \\ n_1 E_{\mathrm{i},\perp} \cos\theta_{\mathrm{i}} - n_1 E_{\mathrm{r},\perp} \cos\theta_{\mathrm{r}} = n_2 E_{\mathrm{t},\perp} \cos\theta_{\mathrm{t}} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} E_{\mathrm{r},\perp} - E_{\mathrm{t},\perp} = -E_{\mathrm{i},\perp} \\ n_1 E_{\mathrm{r},\perp} \cos\theta_{\mathrm{r}} + n_2 E_{\mathrm{t},\perp} \cos\theta_{\mathrm{t}} = n_1 E_{\mathrm{i},\perp} \cos\theta_{\mathrm{i}} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} E_{\mathrm{r},\perp} = \dfrac{-n_2 \cos\theta_{\mathrm{t}} + n_1 \cos\theta_{\mathrm{i}}}{n_2 \cos\theta_{\mathrm{t}} + n_1 \cos\theta_{\mathrm{r}}} E_{\mathrm{i},\perp} \\ E_{\mathrm{t},\perp} = \dfrac{n_1 \cos\theta_{\mathrm{i}} + n_1 \cos\theta_{\mathrm{r}}}{n_2 \cos\theta_{\mathrm{t}} + n_1 \cos\theta_{\mathrm{r}}} E_{\mathrm{i},\perp} \end{cases}$$

$$\begin{cases} -n_1 E_{\mathrm{i},//} + n_1 E_{\mathrm{r},//} = -n_2 E_{\mathrm{t},//} \\ -E_{\mathrm{i},//}\cos\theta_{\mathrm{i}} - E_{\mathrm{r},//}\cos\theta_{\mathrm{r}} = -E_{\mathrm{t},//}\cos\theta_{\mathrm{t}} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} n_1 E_{\mathrm{r},//} + n_2 E_{\mathrm{t},//} = n_1 E_{\mathrm{i},//} \\ E_{\mathrm{r},//}\cos\theta_{\mathrm{r}} - E_{\mathrm{t},//}\cos\theta_{\mathrm{t}} = -E_{\mathrm{i},//}\cos\theta_{\mathrm{i}} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} E_{\mathrm{r},//} = \dfrac{-n_1\cos\theta_{\mathrm{t}} + n_2\cos\theta_{\mathrm{i}}}{-n_1\cos\theta_{\mathrm{t}} - n_2\cos\theta_{\mathrm{r}}} E_{\mathrm{i},//} \\ E_{\mathrm{t},//} = \dfrac{-n_1\cos\theta_{\mathrm{i}} - n_1\cos\theta_{\mathrm{r}}}{-n_1\cos\theta_{\mathrm{t}} - n_2\cos\theta_{\mathrm{r}}} E_{\mathrm{i},//} \end{cases}$$

$$\begin{cases} n = n_2 / n_1 \\ \theta_i = \theta_r \\ n_1 \sin\theta_i = n_2 \sin\theta_t \end{cases} \Rightarrow \sin\theta_i = n \sin\theta_t$$

$$\Rightarrow n\cos\theta_t = \left[n^2 - n^2\sin^2\theta_t\right]^{1/2} = \left[n^2 - \sin^2\theta_i\right]^{1/2}$$

$$\begin{cases} E_{r,\perp} = \dfrac{-n_2\cos\theta_t + n_1\cos\theta_i}{n_2\cos\theta_t + n_1\cos\theta_r} E_{i,\perp} = \dfrac{\cos\theta_i - \left[n^2 - \sin^2\theta_i\right]^{1/2}}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} E_{i,\perp} \\ E_{t,\perp} = \dfrac{n_1\cos\theta_i + n_1\cos\theta_r}{n_2\cos\theta_t + n_1\cos\theta_r} E_{i,\perp} = \dfrac{2\cos\theta_i}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} E_{i,\perp} \end{cases}$$

$$n = \frac{n_2}{n_1}$$

$$n\cos\theta_t = \left[n^2 - \sin^2\theta_i\right]^{1/2}$$

$$\begin{cases} E_{r,//} = \dfrac{-n_1\cos\theta_t + n_2\cos\theta_i}{-n_1\cos\theta_t - n_2\cos\theta_r} E_{i,//} = \dfrac{\left[n^2 - \sin^2\theta_i\right]^{1/2} - n^2\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i} E_{i,//} \\ E_{t,//} = \dfrac{-n_1\cos\theta_i - n_1\cos\theta_r}{-n_1\cos\theta_t - n_2\cos\theta_r} E_{i,//} = \dfrac{2n\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i} E_{i,//} \end{cases}$$

$$E_i = E_{io} expj(\omega t - k_i.r)$$
$$E_r = E_{ro} expj(\omega t - k_r.r)$$
$$E_t = E_{to} expj(\omega t - k_t.r)$$

$$r_s = r_\perp = \frac{E_{r0,\perp}}{E_{i0,\perp}} = \frac{-n_2 \cos\theta_t + n_1 \cos\theta_i}{n_2 \cos\theta_t + n_1 \cos\theta_r} = \frac{\cos\theta_i - \left[n^2 - \sin^2\theta_i\right]^{1/2}}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}}$$

$$t_s = t_\perp = \frac{E_{t0,\perp}}{E_{i0,\perp}} = \frac{n_1 \cos\theta_i + n_1 \cos\theta_r}{n_2 \cos\theta_t + n_1 \cos\theta_r} = \frac{2\cos\theta_i}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}}$$

$$r_p = r_\parallel = \frac{E_{r0,\parallel}}{E_{i0,\parallel}} = \frac{-n_1 \cos\theta_t + n_2 \cos\theta_i}{-n_1 \cos\theta_t - n_2 \cos\theta_r} = \frac{\left[n^2 - \sin^2\theta_i\right]^{1/2} - n^2 \cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2 \cos\theta_i}$$

$$t_p = t_\parallel = \frac{E_{t0,\parallel}}{E_{i0,\parallel}} = \frac{-n_1 \cos\theta_i - n_1 \cos\theta_r}{-n_1 \cos\theta_t - n_2 \cos\theta_r} = \frac{2n\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2 \cos\theta_i}$$

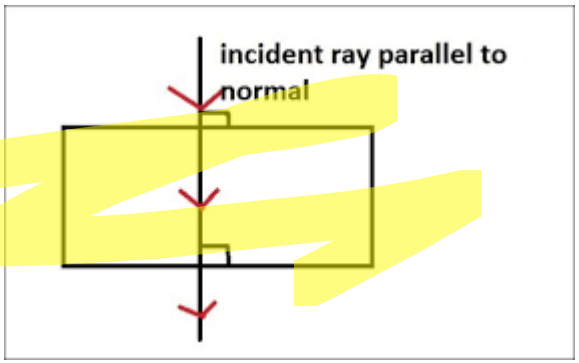
Show that,

$r_p + nt_p = 1$

$r_s + 1 = t_s$

r_s, and t_s, reflection and transmission coefficients for s-polarization r_p, and t_p are the reflection and transmission coefficients for p-polarization.

- When θ_i=0. normal incidence (incident ray parallel to the normal)

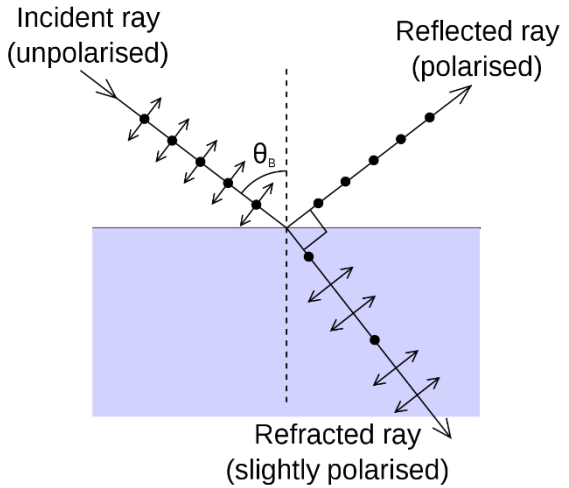


$$r_s = \frac{\cos\theta_i - \left[n^2 - \sin^2\theta_i\right]^{1/2}}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} = \frac{1-n}{1+n} = \frac{n_1 - n_2}{n_1 + n_2}$$

$$t_s = \frac{2\cos\theta_i}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}} = \frac{2}{1+n} = \frac{2n_1}{n_1 + n_2}$$

$$r_p = \frac{\left[n^2 - \sin^2\theta_i\right]^{1/2} - n^2\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i} = \frac{n - n^2}{n + n^2} = \frac{n_1 - n_2}{n_1 + n_2}$$

$$t_p = \frac{2n\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i} = \frac{2n}{n + n^2} = \frac{2n_1}{n_1 + n_2}$$



When r_p=0 reflected light is polarized

$$\left[n^2 - \sin^2\theta_p\right]^{1/2} - n^2\cos\theta_p = 0$$

$$\left[n^2 - \sin^2\theta_p\right]^{1/2} = n^2\cos\theta_p$$

$$n^2 - \sin^2\theta_p = n^4\cos^2\theta_p$$

$$\frac{n^2}{\cos\theta_p} - \frac{\sin^2\theta_p}{\cos^2\theta_p} = n^4$$

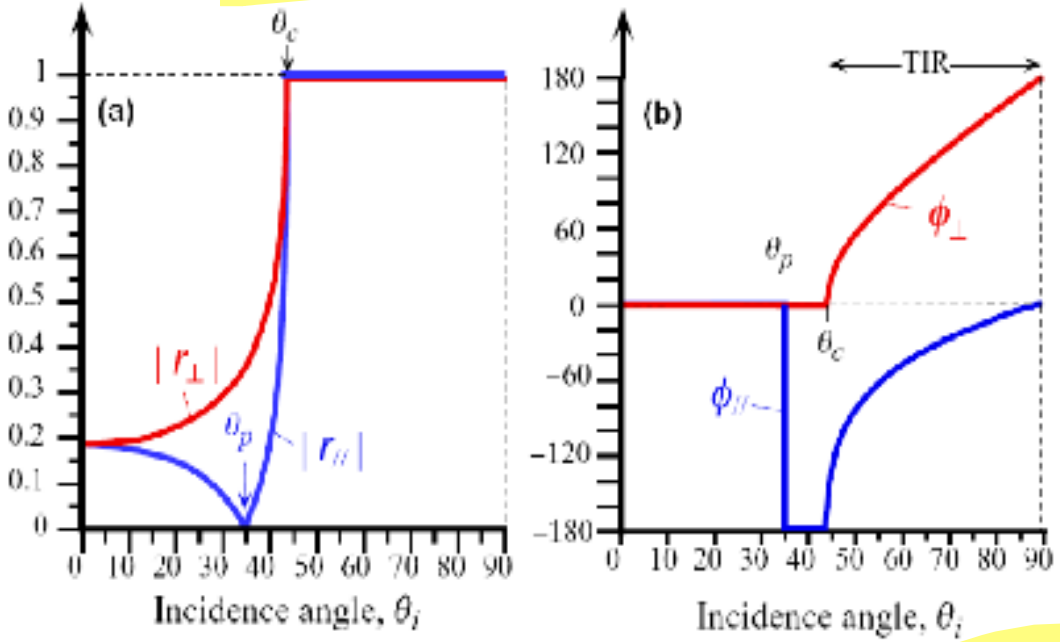
$$n^2(1 + \tan^2\theta_p) - \tan^2\theta_p = n^4$$

$$\tan\theta_p = n = \frac{n_2}{n_1}$$

Here, θ_p is called the polarization angle or Brewster angle.

Brewster's Angle (Polarization Angle)

$n_1 = 1.44, n_2 = 1.00$





David Brewster (1781-1868)

$$\theta_p = 34.78°$$

$$\theta_c = 43.98°$$

- When $\theta_i = \theta_p$, reflected light is linearly polarized and is perpendicular to the incidence plane.
- When $\theta_p < \theta_i < \theta_c$, there is a phase shift of 180° in $r_{//}$.
- $|r_\perp|$ and $|r_{//}|$ increases with θ_i when $\theta_i > \theta_p$.
- There are phase shifts (other than 0 or 180°) in $r_\perp$ and $r_{//}$ when $\theta_i > \theta_c$.

Example: calculate $r_\perp$ and $r_{//}$ when $n_1 = 1.44$, $n_2 = 1.00$, and $\theta_i = 40°$.

Solution:

$$n = \frac{n_2}{n_1} = \frac{1.00}{1.44} = 0.694$$

$$\theta_i < \theta_c = 43.98°$$

$$r_\perp = \frac{\cos\theta_i - \left[n^2 - \sin^2\theta_i\right]^{1/2}}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}}$$

$$= \frac{\cos 40° - \left[0.694^2 - \sin^2 40°\right]^{1/2}}{\cos 40° + \left[0.694^2 - \sin^2 40°\right]^{1/2}} = 0.490$$

$$r_{//} = \frac{\left[n^2 - \sin^2\theta_i\right]^{1/2} - n^2\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i}$$

$$= \frac{\left[0.694^2 - \sin^2 40°\right]^{1/2} - 0.694^2\cos 40°}{\left[0.694^2 \quad \sin^2 40°\right]^{1/2} + 0.694^2\cos 40°} = -0.170$$

Example: calculate $r_\perp$ and $r_{//}$ when $n_1 = 1.00$, $n_2 = 1.44$, and $\theta_i = 40°$.

Solution:

$$n = \frac{n_2}{n_1} = \frac{1.44}{1.00} = 1.44$$

$$r_\perp = \frac{\cos\theta_i - \left[n^2 - \sin^2\theta_i\right]^{1/2}}{\cos\theta_i + \left[n^2 - \sin^2\theta_i\right]^{1/2}}$$

$$= \frac{\cos 40° - \left[1.44^2 - \sin^2 40°\right]^{1/2}}{\cos 40° + \left[1.44^2 - \sin^2 40°\right]^{1/2}} = -0.255$$

$$r_{//} = \frac{\left[n^2 - \sin^2\theta_i\right]^{1/2} - n^2\cos\theta_i}{\left[n^2 - \sin^2\theta_i\right]^{1/2} + n^2\cos\theta_i}$$

$$= \frac{\left[1.44^2 - \sin^2 40°\right]^{1/2} - 1.44^2\cos 40°}{\left[1.44^2 \quad \sin^2 40°\right]^{1/2} + 1.44^2\cos 40°} = -0.104$$

Reflection Coefficients for an Air-to-Glass Interface

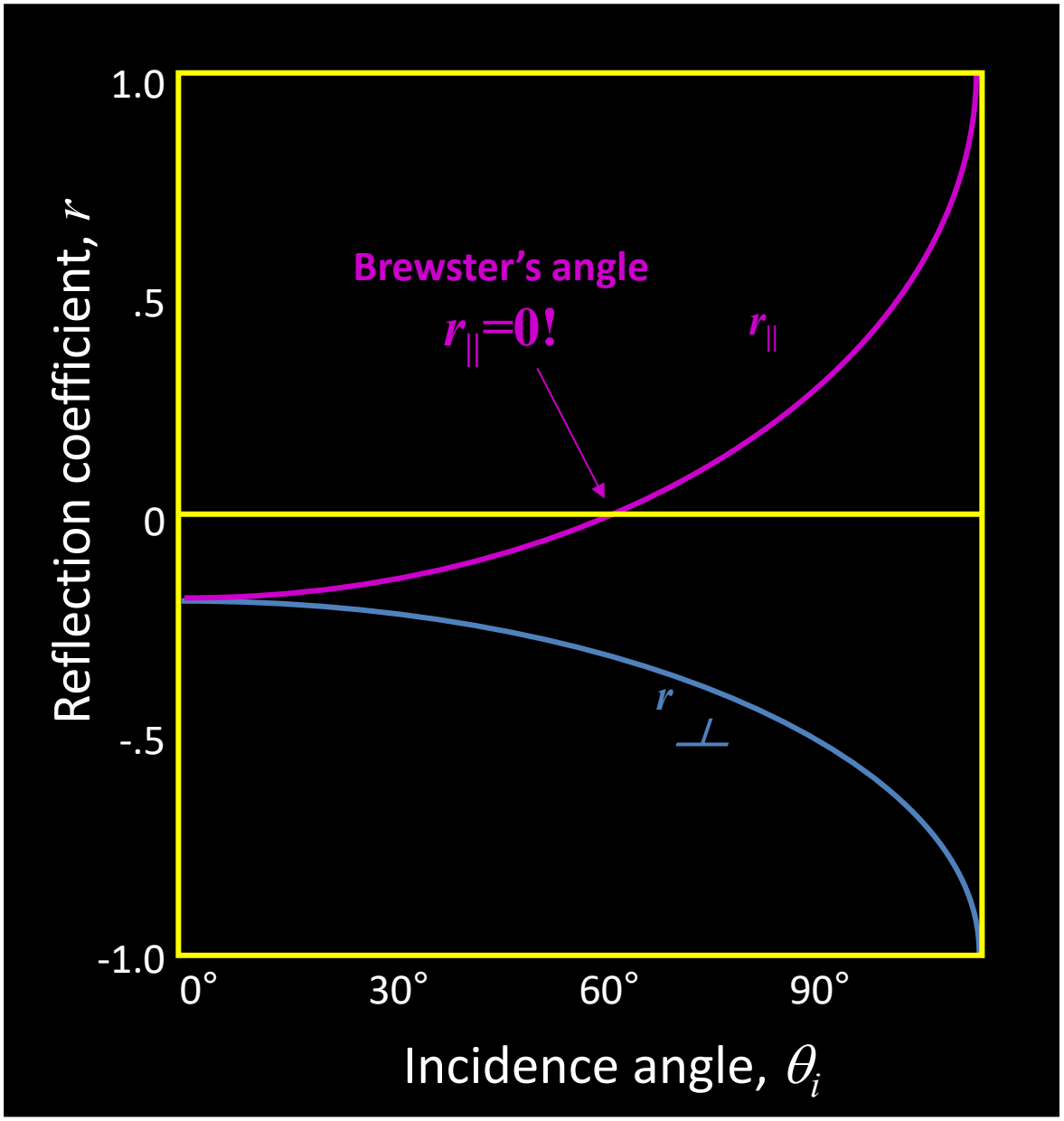
$$n_{air} \approx 1 \; < \; n_{glass} \approx 1.5$$

Note that:

Total reflection at $\theta = 90°$ for both polarizations

Zero reflection for parallel polarization at **Brewster's angle** (56.3° for these values of n_i and n_t).

(We'll delay a derivation of a formula for Brewster's angle until we do dipole emission and polarization.)



Reflection Coefficients for a Glass-to-Air Interface

$n_{glass} \approx 1.5 \; > \; n_{air} \approx 1$

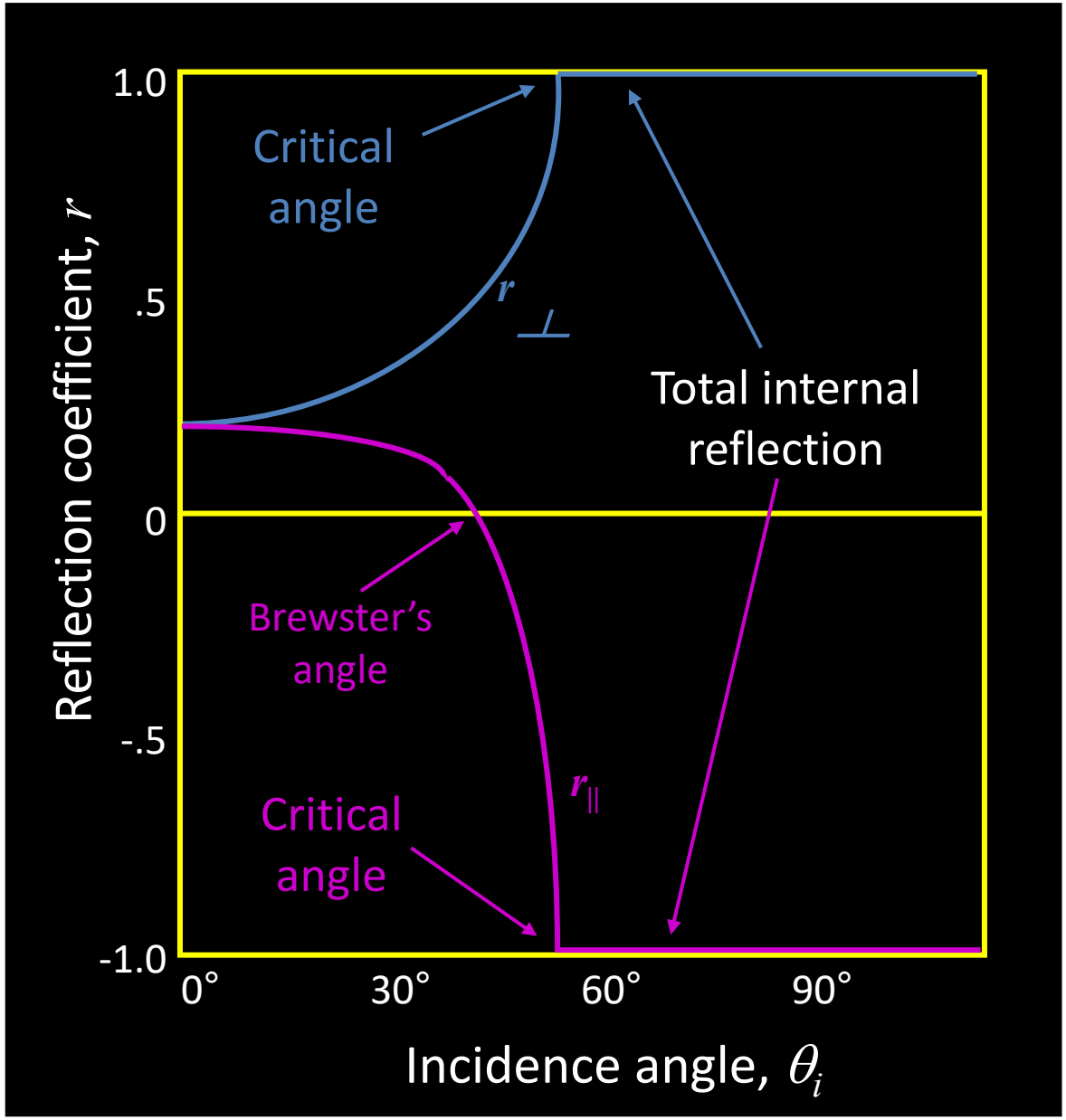
Note that:

Total internal reflection above the **critical angle**

$\theta_{crit} \equiv \arcsin(n_t / n_i)$

(The sine in Snell's Law can't be > 1!):

$\sin(\theta_{crit}) = n_t / n_i \sin(90°)$



Intensity, Reflectance, and Transmittance

Light Intensity, $I = \frac{1}{2} c \varepsilon_r \varepsilon_0 E_0^2 \propto n E_0^2$

Reflectance, $R_s = |r_s|^2$

Reflectance, $R_p = |r_p|^2$

Transmittance, $T_s = |t_s|^2$

Transmittance, $T_p = |t_p|^2$

Example: Consider a light ray at normal incidence on a boundary between a glass medium of refractive index 1.5 and air of refractive index 1.0. (a) If light is traveling from air to glass, what are the reflectance and transmittance? (b) If light is traveling from glass to air, what are the reflectance and transmittance? (c) What is the Brewster's angle when light is traveling from air to glass?

Solution:

$$\begin{cases} n_1 = 1.0 \\ n_2 = 1.5 \end{cases} \Rightarrow \begin{cases} R = \left(\dfrac{n_1 - n_2}{n_1 + n_2}\right)^2 = \left(\dfrac{1.0 - 1.5}{1.0 + 1.5}\right)^2 = 0.04 \\ T = \dfrac{4 n_1 n_2}{(n_1 + n_2)^2} = \dfrac{4 \times 1.0 \times 1.5}{(1.0 + 1.5)^2} = 0.96 \end{cases}$$

$$\begin{cases} n_1 = 1.5 \\ n_2 = 1.0 \end{cases} \Rightarrow \begin{cases} R = \left(\dfrac{n_1 - n_2}{n_1 + n_2}\right)^2 = \left(\dfrac{1.5 - 1.0}{1.5 + 1.0}\right)^2 = 0.04 \\ T = \dfrac{4 n_1 n_2}{(n_1 + n_2)^2} = \dfrac{4 \times 1.5 \times 1.0}{(1.5 + 1.0)^2} = 0.96 \end{cases}$$

$$\tan\theta_p = \frac{n_2}{n_1} = \frac{1.5}{1.0} \Rightarrow \theta_p = 56.31^\circ$$

Light will lose 4% of its intensity whenever it travels through an interface between air and glass.

VVI

Antireflection coatings

Current commercial solar cells are made of silicon (n=3.5 at the wavelengths of 700-800 nm)

Without coatings:

we know, for normal incidence,

$R = \frac{(n_1 - n_2)^2}{(n_1 + n_2)^2} = \frac{(1-3.5)^2}{(1+3.5)^2} = 0.31$, so, light will lose 31% of its intensity when travels from air to solar cells without coating.

Design: When B reaches A, the phase difference= $k_c(2d)$

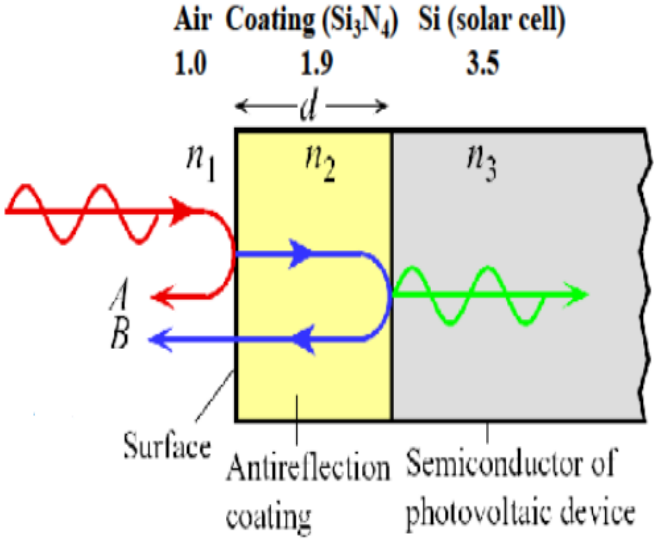
$= \frac{4\pi d}{\lambda_c}$, here λ_c is the wavelength in the coating,

since $\lambda_c = \lambda/n_2$, the phase difference= $\frac{4\pi d n_2}{\lambda}$

To reduce reflectance, A and B must interfere destructively, this requires, the phase difference to be π, or multiple of π, mπ, where m= 1, 3, 5….

$$\frac{4\pi d n_2}{\lambda} = m\pi$$

$$d = m\left(\frac{\lambda}{4n_2}\right)$$



Antireflection coatings

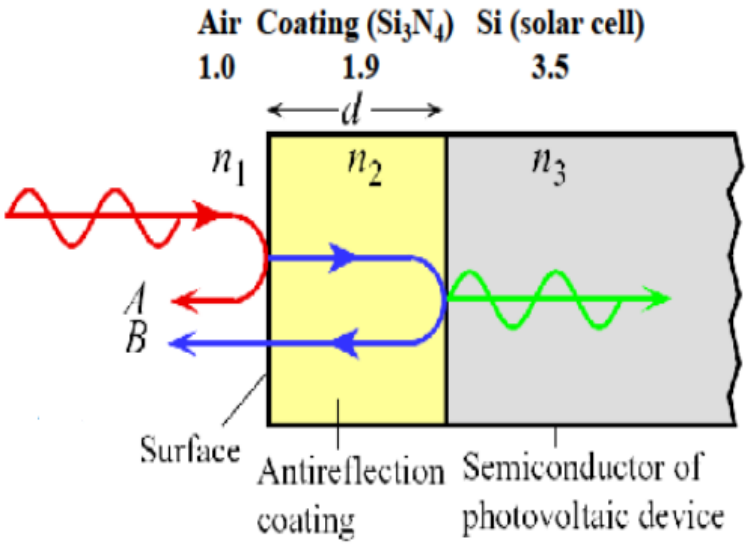
Design: Also, to get good degree of destructive interference between waves A and B, $r_{12} = r_{23}$

$$\frac{n_1 - n_2}{n_1 + n_2} = \frac{n_2 - n_3}{n_2 + n_3}$$

$$\Rightarrow n_2 = \sqrt{n_1 n_3}$$

So, $n_2 = \sqrt{1 \times 3.5} = 1.87$, the closet materials to this refractive index is Silicon nitride, Si_3N_4.

And, $d = \frac{\lambda}{4n_2} = \frac{700nm}{4\times 1.90} = 92\ nm.$



After Coating:

$$R = \left(\frac{n_1 n_3 - {n_2}^2}{n_1 n_3 + {n_2}^2}\right)^2 = \left(\frac{3.8 - (1.9)^2}{3.8 + (1.9)^2}\right)^2 = 0.00065$$

Now, Light will lose only 0.065% of its intensity which is much better compared with 31%

Reading Materials

S. O. Kasap, “Optoelectronics and Photonics: Principles and Practices”, Prentice Hall, Upper Saddle River, NJ 07458, 2001, Chapter 1, “Wave Nature of Light”.